NEUROFY

PortaSENSE BLE communication Specification

Document version: V1.2

Date: January 4, 2026

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Introduction:

This document describes the Bluetooth Low Energy (BLE) communication protocol implemented by the PortaSense device. It provides a complete overview of GATT services, characteristics, security settings, and data exchange formats required for successful communication with a mobile application.

The goal of this document is to enable mobile developers to implement reliable and secure BLE communication into their applications when interacting with the device. It includes technical guidance on how to establish a connection, pair, negotiate MTU, and exchange commands and data packets.

This specification is intended for mobile app developers (iOS/Android) responsible for implementing BLE connectivity, managing security features, and handling communication with the PortaSense device at the GATT level.

Device Overview:

The PortaSense is a custom-designed BLE peripheral device that provides wireless real-time communication with mobile applications. It operates as a BLE 5.0 peripheral and advertises itself under the name "PS_SN1" (PortaSense Serial Number 1) and it can be change.

The device exposes multiple GATT services for configuration, data transmission, and status monitoring. It supports secure connections, bonding, and MTU extension for efficient data exchange.

The PortaSense is designed for low-power operation and can be integrated into various mobile workflows that require real-time control, data collection, or telemetry.

Key specifications include:

- BLE Role: Peripheral
- BLE Version: 5.0
- Supports MITM Protection: Yes
- Preferred MTU Size: 517 Bytes

GATT Services and Characteristics:

This GATT server implementation defines multiple primary services and their corresponding characteristics, structured for a sensor device supporting features such as gas sensing, GNSS, LoRaWAN configuration, OTA updates, system control, alarm reporting, and logging. Below is a breakdown of all services and their roles:

1. **Device Information Service:** Provides basic manufacturer data.
    - Primary UUID: 180A.

Characteristic Name	UUID	Properties	Byte Size	Data Format
Manufacturer Name	2A29	Read (ENC/ AUTHEN)	non-fixed	UTF-8 string
Model Number	2A24	Read (ENC/ AUTHEN)	non-fixed	UTF-8 string
Serial Number	2A25	Read (ENC/ AUTHEN)	non-fixed	UTF-8 string
Hardware Revision	2A27	Read (ENC/ AUTHEN)	non-fixed	UTF-8 string
Firmware Revision	2A28	Read (ENC/ AUTHEN)	non-fixed	UTF-8 string

2. **Measurements Duration Service:** Determining the periodic time of measurement of parameters.
    - Primary UUID: 7437804B-D19B-4749-85DA-A9A74EAD5A91.

Characteristic Name	UUID	Properties	Byte Size	Data Format
Gas & Thermometer	601406A8-303D-47A3-8508-D85FB55FDFD9	Read/Write (ENC/ AUTHEN)	1	uint8
GNSS	C3D82C09-9897-4E17-A1C9-6CA798B7D5D6	Read/Write (ENC/ AUTHEN)	1	uint8
Battery	3F625526-CC10-48C8-A285-4C734DC6C478	Read/Write (ENC/ AUTHEN)	1	uint8

    - These characteristics can be set from 0 to 47, each value demonstrating half an hour, and in total can be set from 30 minutes to 24 hours.

3. **LoRaWAN Configuration service:** Configuring LoRa communication parameter settings.
    - Primary UUID: 83AF025B-E3B6-4707-AF56-F2E404E04DF2

Characteristic Name	UUID	Properties	Byte Size	Data Format
Search Duration	AEB093ED-6A4A-4F64-BCA3-7D94B50CDC44	Read/Write (ENC/ AUTHEN)	1	uint8
Sleep Duration After Success TX	84A9186D-45D8-4150-9C12-9D9338FD551B	Read/Write (ENC/ AUTHEN)	1	uint8
Valid Search Time	91AE609F-E21B-4734-A9CD-60B7F8EB90EB	Read/Write (ENC/ AUTHEN)	4	uint8
Configuration Type	8BCFA435-C596-44D3-BC45-BF1AD5BD0102	Read/Write (ENC/ AUTHEN)	1	uint8
AppEUI	8E507013-4D3C-4139-860C-12E3B78C3612	Read/Write (ENC/ AUTHEN)	8	uint8
DevEUI	1725C1CC-E88C-416D-BCB5-7DC9CEE2BB96	Read/Write (ENC/ AUTHEN)	8	uint8
AppKEY	1BC971EA-F247-4F20-9632-80AD3DB872BA	Read/Write (ENC/ AUTHEN)	16	uint8

- Search Duration and Sleep Duration After Success TX characteristics can be configured similarly part two, with adjustable values ranging from 0 to 47 with 30-minute intervals.
- The Valid Search Time characteristic consists of four bytes representing entry and exit times:
    I. Byte 1: Entry hour (0-23).
    II. Byte 2: Entry minute (0-59).
    III. Byte 3: Exit hour (0-23).
    IV. Byte 3: Exit minute (0-59).
- AppEUI, DevEUI, and AppKEY characteristics are LoRaWAN parameters that accept any hexadecimal values.
- The configuration type determines whether the default values for the AppKEY, AppEUI, and DevEUI parameters are considered or the values set using the mobile app. For default value 0xAA otherwise 0x55.

4. **OTA Service:** The Over the Air (OTA) update mechanism allows a device to update itself.
    - Primary UUID: 884A4483-9A43-4454-BE67-4D64A2FA619F

Characteristic Name	UUID	Properties	Byte Size	Data Format
OTA Control	A783F9E6-049C-4F4E-8E1F-857F3598A32E	Read/Write/NOTIFY (ENC/ AUTHEN)	1	uint8
OTA Data	7D7306E5-5BFE-40AE-A93D-BF9782CE8E4B	Write (ENC/ AUTHEN)	non-fixed	uint8

- The OTA Control characteristic facilitates firmware updates through a structured message exchange:
    I. Update Request (0xA0)
        - Initiated by mobile application to begin OTA process.
        - Requires acknowledgment before proceeding.

    II. Update Acknowledgment (0xA1)
        - Device response confirming readiness for OTA Data transfer.

    III. Transfer Completion (0xB0)
        - Sent by mobile app after final OTA Data packet transmission.
        - Triggers firmware verification and application.

    IV. Update Completion Acknowledgment (0xB1)
        - Device confirmation of successful firmware update.

- Protocol flow:
    I. Mobile app → Device: 0xA0 (Update Request).

    I. Device → Mobile app: 0xA1 for acknowledgment or 0xA2 for negative acknowledgment).

    II. Mobile app → Device: OTA Data packets (via OTA Data characteristic).

    III. Mobile app → Device: 0xB0 (Transfer Complete).

    IV. Device → Mobile app: 0xB1 for update success or 0xB2 for unsuccess.

5. **System Control Service:** Includes settings for various sections.
    - Primary UUID: DCBB6DC3-C47A-48A8-8AF1-B3F7E0A5B45D

Characteristic Name	UUID	Properties	Byte Size	Data Format
Reset to Factory	6BDB5421-D54A-4336-A4A4-79C93AEF0AE5	Write (ENC/ AUTHEN)	1	uint8
BLE Name	03E2251B-20F0-4FCA-B8F1-01640AE0BBE9	Read/Write (ENC/ AUTHEN)	non-fixed	UTF-8 string
BLE Pass Key	996C916B-C858-4729-A404-F87CBEBD6EF2	Read/Write (ENC/ AUTHEN)	4	uint8
Battery Type	5DBC29AF-8E8E-4A22-9CDD-0D1AAA9CC886	Read/Write (ENC/ AUTHEN)	1	uint8
Occupancy Detect Delay	A47C4A33-4C34-46E1-B462-DD938BA6E94E	Read/Write (ENC/ AUTHEN)	1	uint8
Instant Sending Test	4B97E62C-9B92-4132-AD92-248E295BBF19	Write (ENC/ AUTHEN)	1	uint8

- The Reset to Factory characteristic restores device settings to factory defaults when written with value 0xA5.
- The BLE Name characteristic allows users to modify the device's BLE display name.
- The BLE pass key allows users to modify the device’s pass key.
- The Occupancy Detect Delay characteristic configured:
    I. Post-detection delay timing.
    II. False alarm prevention mechanisms.
- Instant Sending Test characteristic:
    I. Force immediate LoRaWAN transmission of data.
- The battery type characteristic allows users to select the battery type from Li-ion with the value 0xAA or LiFePo4 with the value 0x55.

6. **Alarm Service:** This part shows the problems of device.
    - Primary UUID: E5DE34D5-AC14-475A-B8F6-728AB695387F

Characteristic Name	UUID	Properties	Byte Size	Data Format
Alarms Status	71B5FF02-CEB8-425B-A326-35AAE811B3CB	Read (ENC/ AUTHEN/ Notify)	1	uint8

    - The alarm status consists of one byte, each bit representing the status of a sensor, which can be checked for error detection in the event of any problems.
    - Bit 0: Gas sensor status.
    - Bit 1: Thermometer sensor status
    - Bit 2: Accelerometer sensor status.
    - Bit 3 to 7 are reserved.
    - In normal mode, these values were equal to 1 and read 0 in case of problems or failures.